

Intern Management System.

Features, I would suggest for Intern Management System – Ashmit Tiwari

1. User Roles & Access

Admin (To be attached to admin dashboard)

- Manage interns, mentors, departments, and internship programs.
- Manage onboarding/offboarding workflows.
- Review attendance, leave, evaluations, documents, and reports.
- Configure templates, announcements, permissions, and system settings.
- Perform authorized bulk operations.

Mentor (To be designed later)

- View assigned interns and their progress.
- Create and assign tasks.
- Review submissions and work logs.
- Provide feedback and conduct evaluations.
- Approve or reject leave where authorized.
- Monitor attendance, deadlines, blockers, and performance.

Intern

- Manage permitted profile information.
- View internship details, mentor, projects, meetings and tasks.
- Record attendance and submit leave requests.
- Submit daily/weekly work logs.
- Upload and access authorized documents.
- View feedback, evaluations, notifications, and certificates.

2. End-to-End Intern Lifecycle

1. Application / Candidate intake
2. Shortlisting and selection
3. Offer generation and acceptance
4. Onboarding and document verification
5. Account creation and access provisioning
6. Mentor / manager / project assignment
7. Active internship
8. Attendance and leave management
9. Task and project execution
10. Daily/weekly work logs
11. Mentoring and feedback
12. Mid-term evaluation
13. Final evaluation

14. Completion and exit process
15. Certificate and completion letter generation
16. Alumni transition and historical record

Recommended internship statuses:

- APPLIED
- SHORTLISTED
- INTERVIEW
- SELECTED
- OFFERED
- ACCEPTED
- ONBOARDING
- ACTIVE
- ON_HOLD
- COMPLETED
- TERMINATED
- REJECTED

3. Core Functional Modules

3.1 Authentication & Account Management

- Secure login/logout and session management.
- Email verification and password reset where applicable.

3.2 User & Organization Management

- User profiles and role assignment.
- Departments, teams, positions, and reporting relationships.
- Mentor and manager assignment.
- Internship program association.

3.3 Internship Program Management

- Create reusable internship programs.
- Define duration, department, position, required skills, documents, evaluations, and milestones.
- Program-specific onboarding checklists.
- Program-specific evaluation templates.
- Program-level analytics and health metrics.

3.4 Intern Management

- Centralized intern profile.
- Personal, academic, internship, skills, project, attendance, leave, evaluation, and document information.
- Internship ID and unique identifiers.

3.4 Onboarding

- Offer and joining workflow.

- Document collection and verification.
- NDA and policy acknowledgement.
- Account creation and access provisioning.
- Mentor/project assignment.
- Orientation checklist.
- Onboarding completion percentage and pending-action tracking.

3.6 Attendance

- Approved attendance capture workflow.
- Present, absent, leave, holiday & weekend statuses.
- Attendance corrections with authorization.
- Attendance history and monthly reports.
- Low-attendance alerts and anomaly detection.
- Attendance audit trail.

3.7 Leave Management

- Leave types and configurable policies.
- Leave request submission.
- Approval/rejection workflow.
- Supporting document upload.
- Leave balance where applicable.
- Approval timestamps and rejection reasons.
- Controlled modification of approved leave with audit history.

3.8 Project Management

- Create and manage internship projects.
- Assign managers, mentors, interns, milestones, and documents.
- Project status and progress tracking.
- Project-level task and performance visibility.

3.9 Task Management

- Create, assign, prioritize, and track tasks.
- Due dates, estimated hours, actual hours, attachments, and comments.
- Task states: TODO, IN_PROGRESS, IN_REVIEW, CHANGES_REQUESTED, COMPLETED, CANCELLED.
- Submission and review workflow.
- Overdue and deadline-risk alerts.

3.10 Daily / Weekly Work Logs

- Intern work-log submission.
- Work completed, next steps, blockers, and hours.
- Mentor review and comments.
- Historical timeline.

3.11 Evaluation System

- Configurable evaluation templates.
- Every 2 months and final evaluations.
- Category-based scoring.
- Strengths, weaknesses, feedback, improvement plan, and recommendation.
- Self-evaluation and 360-degree feedback as optional workflows.
- Versioned evaluations so historical versions are never silently overwritten.

3.12 Document Management

- Secure upload and storage of resumes, offer letters, NDAs, IDs, certificates, completion letters, and other authorized documents.
- Document metadata and categorization.
- Private access controls.
- Versioning where required.
- Document verification status.
- Signed/private URLs or equivalent secure access mechanism.

3.13 Notifications & Announcements

- Email notifications.
- Task, leave, evaluation, document, onboarding, and internship-end notifications.
- Targeted announcements for all interns, departments, programs, projects, or selected users.
- Read/unread state and notification history.

3.14 Certificates & Completion

- Completion eligibility checks.
- Admin approval workflow.
- Automatic certificate generation.
- Certificate ID generation.
- Secure certificate storage.
- Public verification page with minimal necessary information.

3.15 Offboarding

- Final evaluation.
- Exit interview.
- Asset return tracking where applicable.
- Access revocation.
- Certificate/completion letter issuance.
- Alumni transition.

3.16 Alumni Management

- Convert completed interns to alumni records.
- Preserve internship, project, skills, evaluation, and certificate history.
- Track future employment/conversion where authorized.
- Maintain a searchable historical talent pool.

4. Advanced & Differentiating Features

4.1 Manager Attention Center

- Single screen showing pending approvals, low attendance, overdue tasks, overdue evaluations, upcoming internship endings, and other items requiring action.

4.2 Internship Health Score

- Program-level health score combining attendance, task completion, engagement, evaluation completion, and completion metrics.

4.3 Intern Attention / Risk Signals

- Identify patterns such as repeated absence, missed deadlines, missing work logs, or negative feedback.
- Use as an attention-support tool only; never as an automatic employment decision.

4.4 Activity Timeline

- Chronological record of important internship events such as task completion, leave decisions, feedback, work logs, evaluations, and document events.

4.5 AI Work-Log Summarization

- Summarize long histories of daily/weekly logs into contributions, technologies, milestones, blockers, and outcomes.

4.6 AI Performance Summary

- Combine authorized attendance, tasks, work logs, feedback, evaluations, and skill growth to assist managers in preparing performance summaries.
- Human review and approval must remain mandatory.

4.7 AI Task Recommendations

- Recommend suitable next tasks based on skills, previous work, project needs, and development gaps.
- Mentor remains the decision maker.

4.8 Personalized Skill Development

- Identify skill gaps and suggest learning objectives or progressively harder tasks.

4.9 Controlled Admin Impersonation / View-As

- Allow authorized support administrators to inspect a user's view without requesting the user's credentials.
- Show a prominent impersonation indicator and log every impersonation session.

5. Non-Functional & Production Requirements

Security

- HTTPS everywhere.
- Secure credential and secret management.
- Strong backend authorization; frontend checks are not security controls.
- Input validation and output encoding.

- Protection against SQL injection, XSS, CSRF where applicable, SSRF, insecure file uploads, and common OWASP risks.
- Rate limiting and abuse protection.
- Secure document access.
- Dependency and vulnerability scanning.
- Least-privilege access.

Data Integrity

- Database constraints and foreign keys.
- Unique constraints for identifiers.
- Transactions for multi-step critical operations.
- Idempotency for retryable operations where required.
- No silent overwrites of important records.

Performance

- Pagination for large lists.
- Proper database indexing.
- Caching only where it provides measurable benefit.
- Background jobs for long-running operations.
- Avoid N+1 query patterns.
- Define API response-time targets for critical endpoints.

Reliability

- Automated backups.
- Backup retention policy.
- Tested database restoration.
- Health checks.
- Graceful failure and retry strategies for external services.
- Disaster recovery procedure.

Observability

- Structured application logs.
- Error tracking.
- Metrics and monitoring.
- Request tracing where appropriate.
- Background job monitoring.
- Service health dashboard.

Accessibility & UX

- Responsive UI.
- Keyboard accessibility.
- Clear validation and error states.
- Consistent design system.
- Loading, empty, success, and failure states for every major workflow.
- Confirmation and undo/recovery patterns for destructive actions.

6. Recommended Technical Architecture

Start with a modular monolith unless scale or organizational requirements clearly justify microservices.

- Frontend: React/Next.js.
- Backend: modular API application with clear domain boundaries.
- Database: PostgreSQL.
- Cache / transient state: Redis where justified.
- Object storage: S3-compatible or approved cloud object storage for documents.
- Background jobs: queue + workers for email, certificate generation, document processing, reports, and AI jobs.
- Observability: centralized logs, error tracking, metrics, and health checks.
- CI/CD: automated linting, type checking, tests, build, security scanning, review, and deployment.

Suggested backend domain modules:

- auth
- users
- interns
- programs
- departments
- projects
- tasks
- attendance
- leaves
- worklogs
- feedback
- evaluations
- skills
- documents
- notifications
- certificates
- analytics
- audit

7. High-Level Data Model

- User
- Role
- Permission
- Department
- Intern
- Internship
- InternshipProgram
- MentorAssignment
- Project

- Task
- TaskSubmission
- Attendance
- LeaveRequest
- WorkLog
- Feedback
- EvaluationTemplate
- Evaluation
- Skill
- InternSkill
- Document
- Notification
- Announcement
- Certificate
- AuditLog

9. API & Backend Standards

- Consistent REST/HTTP semantics or another explicitly documented API style.
- Authentication and authorization middleware on protected endpoints.
- Schema validation for every external input.
- Consistent success/error response structure.
- Stable error codes for client handling.
- Pagination, filtering, sorting, and search conventions standardized across APIs.
- API versioning strategy.
- OpenAPI/Swagger documentation.
- Idempotency for operations that may be retried.
- No stack traces, secrets, tokens, or internal database details in client-facing errors.

10. Testing Strategy

- Unit tests for domain/business logic.
- Integration tests for database and service interactions.
- API tests for authentication, authorization, validation, and critical workflows.
- End-to-end tests for complete lifecycle workflows.
- Permission/RBAC tests for every sensitive module.
- File-upload and document-access tests.
- Background-job tests.
- Regression test suite before releases.
- Security testing and dependency scanning.
- Load/performance testing for expected production traffic..

11. Recommended Delivery Roadmap

Phase 1 — Foundation

- Authentication
- RBAC
- Users
- Departments
- Interns
- Programs
- Audit Logs
- Database foundation

Phase 2 — Core Operations

- Attendance
- Leave
- Projects
- Tasks
- Work Logs
- Documents
- Notifications

Phase 3 — Performance

- Feedback
- Evaluations
- Performance Dashboards
- Manager Attention Center

Phase 4 — Lifecycle

- Onboarding
- Offboarding
- Certificates
- Certificate Verification
- Alumni

Phase 5 — Production Hardening

- Automated testing
- Security review
- Monitoring
- Backups
- CI/CD
- Load testing
- Disaster recovery